

Curriculum Vitae

Education

Nov 2021 - OCT 2024	P.h.D. in COMPUTER SCIENCE, ROBOTICS AND GEOINFORMATION, Grade: top 5% (Outstanding) University of Rome Tor Vergata <i>Thesis Title: Innovative Solutions for Speculative Parallel Discrete Event Simulation on Multi-core Shared-memory Machines</i> Supervisor: Francesco QUAGLIA, Co-Supervisor: Romolo MAROTTA
MAY 2018 - OCT 2021	Master Degree in COMPUTER SCIENCE ENGINEERING University of Rome Tor Vergata Grade: 104/110 Thesis Title: Studio e implementazione di uno stack sequentially consistent Advisor: Francesco QUAGLIA, Co-Advisor: Romolo MAROTTA
OCT 2013 - APR 2018	Bachelor Degree in COMPUTER SCIENCE ENGINEERING Grade: 104/110 University of Rome Tor Vergata
SEPT 2008 - JULY 2013	Liceo Classico Claudio Eliano , Palestrina Grade: 85/100

Research Activities

MARCH 2026–MARCH 2027	Researcher (Incarico di ricerca, art. 22-ter L. 240/2010) Horizon Europe Project <i>EoCoE-3: Energy Oriented Centre of Excellence for Exascale</i> (Grant Agreement : 101144014) Department of Civil Engineering and Computer Science – Università Roma Tor Vergata
NOV 2025–MARCH 2026	Scientific collaborator PRIN 2022 Project <i>DOMAIN: Taming Heterogeneous Computing Complexity with Full-Stack Governance of Domain-Specific Languages</i> (Project Code : 2022TSYYKJ) Department of Economics, Università Roma Tre
NOV 2024–NOV 2025	Research Fellow (Assegno di ricerca, art. 22 L. 240/2010) PRIN 2022 Project <i>DOMAIN: Taming Heterogeneous Computing Complexity with Full-Stack Governance of Domain-Specific Languages</i> (Project Code : 2022TSYYKJ) National Council of Research (CNR-IASI)

Summary of Research Results (Dec 2025)

Bibliometric Index	Scopus
Total Number of Publications	13
Hirsch Index (h-index)	4
Normalized Hirsch Index (m-index)	1
Total Number of Citations	29
Average Citations per Publication	2.23
Total Impact Factor	3.8
Average Impact Factor	1.9

Teaching Activities

NOVEMBER 2023–JANUARY 2024	Assistant Collaborator/Tutor at University of Rome Tor Vergata for the Computer Engineering Bachelor Degree's course Fundamentals of Computer Science
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National/International Research Projects

MARCH 2026–MARCH 2027	EoCoE3: Centro di Eccellenza per l'Energia , <i>Grant Agreement 101144014 – Department of Civil Engineering and Computer Science – Università Roma Tor Vergata</i> , Researcher Research activity within EoCoE-3 (Horizon Europe), focusing on the development and optimization of HPC libraries in PSCToolkit for exascale architectures, and their integration into applications for hydrological and wind energy simulations, with emphasis on accelerator-based implementations.
NOVEMBER 2024–MARCH 2026	DOMAIN , <i>Taming Heterogeneous Computing Complexity with Full-Stack Governance of Domain-Specific Languages</i> , <i>PRIN 2022, CUP F53D23004370001</i> , Research Fellow Research on heterogeneous computing systems and domain-specific languages (DSLs), focusing on programming abstractions, runtime systems, and optimization techniques for efficient execution on hybrid architectures (CPU/GPU/FPGA).

Awards

2025	Best Paper Award , 30TH IEEE INTERNATIONAL SYMPOSIUM ON DISTRIBUTED SIMULATION AND REAL TIME APPLICATIONS (DS-RT) Award for the paper <i>Model-Driven Parallel and Distributed Stochastic Simulation of Chemical Reaction Networks</i>
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Memberships

01 MAY 2025-ONGOING	ACM SIGHPC Association for Computing Machinery's Special Interest Group on High Performance Computing, Member
JUNE 2022-ONGOING	ACM Association of Computer Machinery, Member

Skills and Knowledge

BASIC KNOWLEDGE:	Operating Systems: Windows 9x, Windows NT, Windows 11 Languages: R, Go, Javascript, Assembly MIPS. IDE: JetBrains MPS Other: Matlab, Docker, AngularJS, Apache Kafka, KafkaStreams, Postman, Apache Flink, Apache Spark, Discrete Mathematics, Boolean Algebra, Reinforcement Learning.
INTERMEDIATE KNOWLEDGE:	Operating Systems: Ubuntu > 14.04 Languages: C++, Python, Assembly x86, SQL IDE: Eclipse, JetBrains Clion, JetBrains IntelliJ, JetBrains PyCharm. Other: Linux, Shell scripting, Git, Profiling Tools (perf), gdb debugger, Database management (MySQL, PostgreSQL), Data analysis (Python's pandas, R), Cloud Computing (Amazon AWS Cloud Services), $\text{\LaTeX}$, Software Modelling (UML), Probability and Statistics, Linear Algebra.
ADVANCED KNOWLEDGE:	Languages: Java, C, Makefile. Other: cmake, Procedural Programming, Object-Oriented Programming, SDLC (Waterfall, Agile), Concurrent and Parallel Systems.
SOFT SKILLS:	Critical-thinking, Curiosity, Collaboration, Commitment, Reliability, Meticulousness, Collaboration, Adaptability

Scientific Activities

Chairmanship

- 2026 **Reproducibility Chair**, *ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*
- 2025 **Reproducibility Chair**, *ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*

Research Grants

- 14-20 JULY 2024 Grant Recipient for the **20th International Summer School on Advanced Computer Architecture and Compilation for High-performance Embedded Systems**, Fiuggi, Italy
- 24-26 JUNE 2024 ACM Grant Recipient for **38th ACM SIGSIM Principles of Advanced Discrete Simulation** conference, Atlanta, Georgia, USA
- 05-07 JUNE 2024 ACM Grant Recipient for **33rd International Symposium on High-Performance Parallel and Distributed Computing**, Pisa, Italy
- 20-22 SEPTEMBER 2023 ACM Scholarship Awardee for **10th ACM WomENCourage**, Trondheim, Norway
- 10-16 JULY 2022 Grant Recipient for the **18th International Summer School on Advanced Computer Architecture and Compilation for High-performance Embedded Systems**, Fiuggi, Italy

Languages

- ENGLISH: *Reading: B2, Listening: B2, Writing: B2, Speaking: B2*
- ITALIAN: First Language

Language Certificates

- MAY 2011 Cambridge FCE B2, **Cambridge institute**
- APR 2010 Cambridge PET B1, **Cambridge institute**

Publications

Journal

- 2025 Montesano: **RCR Report for the Paper: “Using (Not-so) Large Language Models to Generate Simulation Models in a Formal DSL: A Study on Reaction Networks”**. ACM TRANSACTIONS ON MODELING AND COMPUTER SIMULATION (SEPTEMBER 2025), IF: 1.9 (YEAR OF REFERENCE 2024)
- 2024 Montesano, Marotta, Quaglia: **Spatial/Temporal Locality-based Load-sharing in Speculative Discrete Event Simulation on Multi-core Machines**, ACM TRANSACTIONS ON MODELING AND COMPUTER SIMULATION 35.1 (2024): 1-31, IF: 1.9

Conference

- 2025 De Angelis, E., De Angelis, G., Marotta, Montesano, Pellegrini, Proietti: **Declarative Adaptive Optimization of Task-Based Applications on Heterogeneous Architectures**, IEEE/SBC 37TH INTERNATIONAL SYMPOSIUM ON COMPUTER ARCHITECTURE AND HIGH PERFORMANCE COMPUTING WORKSHOPS (SBAC-PADW), 2025
- 2025 Bauco, Montesano, Pimpini, Marotta, Pellegrini: **Model-Driven Parallel and Distributed Stochastic Simulation of Chemical Reaction Networks.**, 29TH INTERNATIONAL SYMPOSIUM ON DISTRIBUTED SIMULATION AND REAL TIME APPLICATIONS (DS-RT). 2025
- 2025 Montesano: **Reproducibility Report for the Paper:“Improving the Efficiency of Coordinating Timed Events in Distributed Systems”**, 39TH ACM SIGSIM INTERNATIONAL CONFERENCE ON PRINCIPLES OF ADVANCED DISCRETE SIMULATION. 2025
- 2024 Montesano, Marotta, Quaglia: **Lightweight Operating System Services for Incremental Checkpointing in Speculative Discrete Event Simulation on Linux Platforms**, 2024 IEEE 30TH INTERNATIONAL CONFERENCE ON PARALLEL AND DISTRIBUTED SYSTEMS (ICPADS). IEEE, 2024
- 2024 Montesano: **Towards the Improvement of Memory Management of PDES Systems in Multi-core Shared-memory Machines** , 2024 28TH INTERNATIONAL SYMPOSIUM ON DISTRIBUTED SIMULATION AND REAL TIME APPLICATIONS (DS-RT). IEEE, 2024.
- 2024 Montesano: **Full-Stack Revision of Memory and Data Management in PDES on Multi-Core Machines**, PROCEEDINGS OF THE 33RD INTERNATIONAL SYMPOSIUM ON HIGH-PERFORMANCE PARALLEL AND DISTRIBUTED COMPUTING. 2024
- 2024 Montesano: **Towards the Optimization of Memory and Data Management in Speculative PDES on Multi-Core Machines**, PROCEEDINGS OF THE 38TH ACM SIGSIM INTERNATIONAL CONFERENCE ON PRINCIPLES OF ADVANCED DISCRETE SIMULATION. 2024
- 2024 Montesano, Wilsdorf: **Reproducibility Report for the Paper: Modeling Central Bank Digital Currency over Payment Channels: A Parallel ROSS-based Approach**, PROCEEDINGS OF THE 38TH ACM SIGSIM INTERNATIONAL CONFERENCE ON PRINCIPLES OF ADVANCED DISCRETE SIMULATION. 2024
- 2023 Marotta, Montesano, Pellegrini, Quaglia: **Incremental Checkpointing of Large State Simulation Models with Write-Intensive Events via Memory Update Correlation on Buddy Pages**, 2023 27TH INTERNATIONAL SYMPOSIUM ON DISTRIBUTED SIMULATION AND REAL TIME APPLICATIONS (DS-RT). IEEE, 2023.
- 2023 Montesano: **Towards the Optimization of Speculative PDES Platforms in Shared-Memory Multi-core Machines**, EUROPEAN CONFERENCE ON MULTI-AGENT SYSTEMS. CHAM: SPRINGER NATURE SWITZERLAND, 2023
- 2023 Marotta, Montesano, Quaglia: **Effective Access to the Committed Global State in Speculative Parallel Discrete Event Simulation on Multi-core Machines**, SIGSIM-PADS 2023
- 2021 Montesano, Marotta, Quaglia: **Spatial/Temporal Locality-based Load-sharing in Speculative Discrete Event Simulation on Multi-core Machines**, SIGSIM-PADS 2022

International Conferences/Workshops/Events

10-14 OCTOBER 2024	Speaker at the 30th International Conference on Parallel and Distributed Systems (ICPADS) , Belgrade, Serbia Paper Title: <i>Lightweight Operating System Services for Incremental Checkpointing in Speculative Discrete Event Simulation on Linux Platforms</i>
07-09 OCTOBER 2024	Speaker (Doctoral Colloquium) at the 28th International Symposium on Distributed Simulation and Real Time Applications (DS-RT) , Urbino, Italy Paper Title: <i>Towards the Improvement of Memory Management in PDES Systems on Multi-core Shared-memory Machines</i>
14-20 JULY 2024	Attendance and Poster presentation at the 20th International Summer School on Advanced Computer Architecture and Compilation for High-performance Embedded Systems , Fiuggi, Italy Poster Title: <i>Improving Memory Awareness in PDES on Multi-Core Shared-Memory Machines</i>
24-26 JUNE 2024	Speaker (PhD Colloquium) at the 38th ACM SIGSIM Principles of Advanced Discrete Simulation Conference , Atlanta, Georgia, USA Paper Title: <i>Towards the Optimization of Memory and Data Management in Speculative PDES on Multi-Core Machines</i>
05-07 JUNE 2024	Speaker (PhD Symposium) at the 33rd International Symposium on High-Performance Parallel and Distributed Computing , Pisa, Italy Paper Title: <i>Full-Stack Revision of Memory and Data Management in PDES on Multi-Core Machines</i>
20-22 SEPTEMBER 2023	Attendance and Hackathon participation at 10th ACM WomENCourage , Trondheim, Norway
13-15 SEPTEMBER 2023	Attendance and Poster presentation at the 20th European Conference on Multi-Agent Systems , Napoli, Italy Poster Title: <i>Towards the Optimization of Speculative PDES Platforms in Shared-Memory Multi-Core Machines</i>
10-16 JULY 2022	Attendance and Poster presentation at the 18th International Summer School on Advanced Computer Architecture and Compilation for High-performance Embedded Systems , Fiuggi, Italy Poster Title: <i>Tackling Memory Locality in Multi-Core Architectures</i>
8-10 JUNE 2022	Attendance (Virtual) at the 36th ACM SIGSIM Principles of Advanced Discrete Simulation Conference Paper Title: <i>Spatial/Temporal Locality-based Load-sharing in Speculative Discrete Event Simulation on Multi-core Machines</i>

Other Activities

20-24 OCTOBER 2025	European First Aid Certificate for the Attendance to First Aid Course , at C.R.I. COMITATO MUNICIPI 8-11-12 DI ROMA ODV,
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Consent to the use of Personal Data

I hereby authorize the use of my personal information in relation to the Italian Legislative Decree n. 196/2003, “Codice in materia di protezione dei dati personali” and in relation to the EU regulation GDPR (2016/679). Everything here declared is the truth compliant to the articles 46 and 47 n. 445 of D.P.R.